Chapter 3. Sequences of Random Variables

Coin tossing, laws of large numbers, independence, product measures, weak convergence, characteristic functions, the central limit theorem

Contents

3.1 Infinitely many fair coin tosses **2**

3.2 Infinitely many biased coin tosses **3**

3.3 The (strong and the weak) law of large numbers **3**

3.4 Independence **6**

3.4.1 Independent events 6

3.4.2 Independent random variables 8

3.5 Kolmogorov's law of large numbers **9**

3.6 Finite products of probability measures **10**

3.7 Weak convergence of probability measures **13**

3.8 Joint distribution and convolution **15**

3.9 Characteristic functions **18**

3.10 Central limit theorem **20**

3.1 Infinitely many fair coin tosses

Motivation. Finitely many coin tosses need only counting. To talk about limits (relative frequencies, waiting times, "infinitely many heads"), all tosses must live on one probability space. We build it from Lebesgue measure on $[0,1]$ via binary expansions, which also shows why uncountable sample spaces force us into measure theory.

A **random experiment** is one whose outcome is known only after it is performed; Ω is the set of possible outcomes (finite, countable or uncountable), and probability attaches a number to every event.

One toss. $\Omega = \{0,1\}$, $\mathcal{A} = \mathcal{P}(\Omega)$, $P(\{0\}) = P(\{1\}) = \frac{1}{2}$.

n tosses. $\Omega = \{0,1\}^n$, $\mathcal{A} = \mathcal{P}(\Omega)$ and $P(A) = |A|/2^n$: the **uniform distribution**, each binary word having probability 2^{-n}.

Infinitely many tosses. $\Omega = \{0,1\}^{\mathbb{N}} = \{(x_i)_{i\in\mathbb{N}} \mid x_i \in \{0,1\}\}$ is uncountable; $(x_i) \mapsto \sum_i x_i 2^{-i}$ maps it onto $[0,1]$ (not injectively). Let $\mathcal{A} := \sigma(\mathcal{A}_0)$ with the *cylinder sets*

$$\mathcal{A}_0 := \{B \subset \Omega \mid B = B_0 \times \{0,1\} \times \{0,1\} \times \cdots \text{ for some } n \text{ and } B_0 \subset \{0,1\}^n\},$$

the events that depend on finitely many tosses. We want $P(x_1 = \bar{x}_1, \dots, x_n = \bar{x}_n) = 2^{-n}$ and must show that such a P extends to $\mathcal{A}$.

Existence via Lebesgue measure. Let $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P}) = ([0,1], \mathcal{B}([0,1]), \lambda)$. On $\Omega = \{0,1\}^{\mathbb{N}}$ let $X_i((x_n)_n) := x_i$ (the i-th toss) and $\mathcal{A} := \sigma(\{X_i = 1\} \mid i \in \mathbb{N})$, which equals $\sigma(\mathcal{A}_0)$. The non-terminating binary expansion of $\tilde{\omega} \in [0,1]$ is

$$\tilde{\omega} = \sum_{i=1}^{\infty} T_i(\tilde{\omega}) 2^{-i}, \qquad T_i := \sum_{k=1}^{2^{i-1}} \mathbb{1}_{\left(\frac{2k-1}{2^i}, \frac{2k}{2^i}\right]},$$

and defines $T : \tilde{\Omega} \to \Omega$, $T(\tilde{\omega}) := (T_1(\tilde{\omega}), T_2(\tilde{\omega}), \dots)$. T is not surjective: sequences ending in infinitely many zeros (other than $(0,0,\dots)$) are missed; they correspond to the dyadic rationals in $(0,1]$, a countable set.

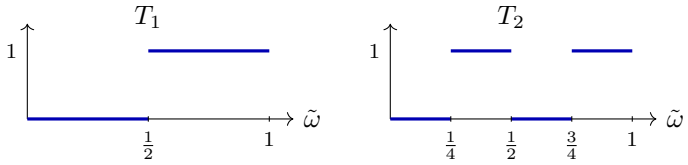


Figure. *The binary digit maps T_1, T_2 (and similarly $T_3, T_4, \dots$). Each $\{T_i = 1\}$ is a finite union of intervals of total length $\frac{1}{2}$, and fixing n digits pins $\tilde{\omega}$ down to an interval of length 2^{-n}.*

Since $T_i = X_i \circ T$, $T^{-1}(\{X_i = 1\}) = \{T_i = 1\}$ is a finite union of intervals, so T is $\tilde{\mathcal{A}}/\mathcal{A}$-measurable (Remark 1.32). $P := \tilde{P} \circ T^{-1}$ is a probability measure on $(\Omega, \mathcal{A})$ (Proposition 2.13), and for $(x_1, \dots, x_n) \in \{0,1\}^n$

$$P(X_1 = x_1, \dots, X_n = x_n) = \tilde{P}\Big(\bigcap_{i \le n} \{T_i = x_i\}\Big) = \tilde{P}(\text{an interval of length } 2^{-n}) = 2^{-n}.$$

So P extends the model for n tosses; uniqueness follows from Theorem 1.9, since the cylinder sets form a $\cap$-stable generator.

Some expectations.

- The i-th toss: $\mathbb{E}[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = \frac{1}{2}$.

- Number of successes $S_n := X_1 + \cdots + X_n$: $P(S_n = k) = \binom{n}{k}2^{-n}$, so $\mathbb{E}[S_n] = \sum_k k\binom{n}{k}2^{-n} = \frac{n}{2}$; more simply, $\mathbb{E}[S_n] = \sum_i \mathbb{E}[X_i] = \frac{n}{2}$ by linearity.
- Waiting time $S := \min\{n \mid X_n = 1\}$ ($\min\emptyset := \infty$), measurable since $\{S = k\} = \{X_1 = \cdots = X_{k-1} = 0, X_k = 1\}$: $P(S = k) = 2^{-k}$, $P(S = \infty) = 0$, and

$$\mathbb{E}[S] = \sum_{k\geq 1} k2^{-k} = \frac{1}{2}\sum_{k\geq 1} k\Big(\frac{1}{2}\Big)^{k-1} = 2, \qquad \text{using } \sum_{k\geq 1} kq^{k-1} = \frac{d}{dq}\frac{q}{1-q} = \frac{1}{(1-q)^2}.$$

- (*Doubling strategy*) A player bets on a fair coin, getting twice the stake on success, and doubles the stake until the first success, starting from 1 unit. The stake in round n is $I_n = 2^{n-1}\mathbb{1}_{\{T>n-1\}}$ with T the waiting time for the first 1. Then $\mathbb{E}[I_n] = 2^{n-1}P(T > n-1) = 2^{n-1}2^{-(n-1)} = 1$ for all n, while $I_n \to 0$ a.s. (for every $\omega \neq (0, 0, \dots)$). Expectations need not follow pointwise limits.

3.2 Infinitely many biased coin tosses

Motivation. Changing the success probability from $\frac{1}{2}$ to p gives the general Bernoulli scheme. Its moments are computed here from the joint law alone; existence and uniqueness of the measure follow from Section 3.6.

With $\Omega = \{0,1\}^{\mathbb{N}}$, X_i and $\mathcal{A}$ as before, there is a unique probability measure $P = P_p$ on $(\Omega, \mathcal{A})$ with

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i} \qquad (1 \leq i_1 < \cdots < i_n).$$

(Existence for $p = \frac{1}{2}$ was shown above; the general case and uniqueness come in Section 3.6 and Example 3.13.) Then $P_p(X_i = 1) = p$ and $P_p(X_i = X_j = 1) = p^2$ for $i \neq j$, so

$$\mathbb{E}_p[X_i] = p, \qquad \mathrm{Var}(X_i) = p - p^2 = p(1-p), \qquad \mathrm{Cov}(X_i, X_j) = \mathbb{E}_p[X_iX_j] - p^2 = 0 \ (i \neq j).$$

The X_i are pairwise uncorrelated (in fact independent, Example 3.13). For $S_n = X_1 + \cdots + X_n$: $\mathbb{E}_p[S_n] = np$, $\mathrm{Var}(S_n) = np(1-p)$ (Bienaymé).

- $X := \sum_n 2^{-n}X_n$ has $\mathbb{E}_p[X] = \sum_n 2^{-n}p = p$ by Theorem 1.56. The partial sums are Cauchy in L^2, and uncorrelatedness gives $\mathrm{Var}(X) = \sum_n 4^{-n}p(1-p) = \frac{1}{3}p(1-p)$.
- The waiting time S for the first success is **geometric**: $P_p(S = n) = (1-p)^{n-1}p$. Hence $\mathbb{E}_p[S] = \sum_n n(1-p)^{n-1}p = \frac{1}{p}$ and $\mathrm{Var}(S) = \frac{1-p}{p^2}$. [Supplement] For the variance, $\mathbb{E}_p[S(S-1)] = p(1-p)\sum_n n(n-1)(1-p)^{n-2} = \frac{2(1-p)}{p^2}$ (second derivative of the geometric series), so $\mathrm{Var}(S) = \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}$.

3.3 The (strong and the weak) law of large numbers

Motivation. Repeating an experiment and averaging should remove the randomness: the empirical mean S_n/n should approach the true mean. Under uncorrelatedness and bounded variances this holds in L^2, in probability (weak law) and almost surely (strong law); the passage to a.s. convergence is the subsequence trick plus Borel–Cantelli.

Throughout, $(\Omega, \mathcal{A}, P)$ is a probability space and $X_1, X_2, \dots \in \mathcal{L}^2$ are uncorrelated ($\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$) with uniformly bounded variances $\sup_i \sigma_i^2 < \infty$, $\sigma_i^2 := \mathrm{Var}(X_i)$. Let $S_n := X_1 + \cdots + X_n$; $S_n(\omega)/n$ is the *empirical mean* of the first n observations. We want $\frac{S_n}{n} \approx \frac{\mathbb{E}[S_n]}{n}$ (resp. $\approx m$ if $\mathbb{E}[X_i] \equiv m$) for large n.

Remark 3.1

We may assume $\mathbb{E}[X_i] = 0$: the centred $\tilde{X}_i := X_i - \mathbb{E}[X_i]$ are in $\mathcal{L}^2$, uncorrelated, have the same variances, and $\frac{\tilde{S}_n}{n} - \frac{\mathbb{E}[\tilde{S}_n]}{n} = \frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}$.

Theorem 3.2 (L^2 law of large numbers)

$\lim_n \mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = 0$ (resp. $\mathbb{E}[(\frac{S_n}{n} - m)^2] \to 0$ if $\mathbb{E}[X_i] \equiv m$).

Proof. By Bienaymé (Remark 2.26),

$$\mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = \operatorname{Var}\Big(\frac{S_n}{n}\Big) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma_i^2 \le \frac{1}{n}\sup_i \sigma_i^2 \to 0.$$

■

Remark 3.3

This is a Hilbert space fact: in L^2 with $(X, Y) = \mathbb{E}[XY]$, the arithmetic mean of uniformly bounded orthogonal vectors tends to 0, since $\|S_n/n\|^2 = n^{-2}\sum_{i\le n}\|X_i\|^2 \to 0$.

Theorem 3.4 (Weak law of large numbers)

If $X_i \in \mathcal{L}^2$ are uncorrelated with uniformly bounded variances and $\mathbb{E}[X_i] = m$, then for every $\varepsilon > 0$, $\lim_n P(|\frac{S_n}{n} - m| \ge \varepsilon) = 0$, i.e. $S_n/n \to m$ in probability.

Proof. Chebyshev: $P(|\frac{S_n}{n} - m| \ge \varepsilon) \le \varepsilon^{-2}\operatorname{Var}(\frac{S_n}{n}) \to 0$ by Theorem 3.2. ■

Example 3.5 (Bernoulli)

For Bernoulli experiments with parameter p, $\mathbb{E}_p[X_i] = p$ and $\operatorname{Var}(X_i) = p(1-p) \le \frac{1}{4}$, so

$$P_p\Big(\Big|\frac{S_n}{n} - p\Big| \ge \varepsilon\Big) \to 0. \tag{3.1}$$

The relative frequency of "1" approaches p (J. Bernoulli, *Ars Conjectandi*), which justifies reading p as a long-run relative frequency.

Example 3.6 (Bernstein polynomials)

Every $f \in C([0,1])$ is a uniform limit of polynomials. For $p \in [0,1]$ the transformation theorem (Remark 2.15) gives

$$B_n(p) := \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k}p^k(1-p)^{n-k} = \sum_k f\Big(\frac{k}{n}\Big)P_p(S_n = k) = \mathbb{E}_p\Big[f\Big(\frac{S_n}{n}\Big)\Big].$$

Given $\varepsilon > 0$, uniform continuity gives $\delta > 0$ with $|f(x) - f(y)| \le \varepsilon$ for $|x - y| \le \delta$. Splitting on $\{|\frac{S_n}{n} - p| \le \delta\}$ and using Chebyshev with $p(1-p) \le \frac{1}{4}$:

$$|B_n(p) - f(p)| \le \mathbb{E}_p\Big|f\Big(\frac{S_n}{n}\Big) - f(p)\Big| \le \varepsilon + 2\|f\|_\infty P_p\Big(\Big|\frac{S_n}{n} - p\Big| > \delta\Big) \le \varepsilon + \frac{2\|f\|_\infty}{4\delta^2 n}.$$

Hence $\limsup_n \|B_n - f\|_\infty \le \varepsilon$ for every ε, i.e. $\|B_n - f\|_\infty \to 0$.

From convergence in probability we now pass to almost sure convergence.

Theorem 3.7 (Strong law of large numbers)

Let $X_1, X_2, \dots \in \mathcal{L}^2$ be uncorrelated with $\sup_i \sigma^2(X_i) = c < \infty$. Then $\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n} \to 0$ P-a.s. (resp. $\frac{S_n}{n} \to m$ a.s. if $\mathbb{E}[X_i] \equiv m$).

Proof. By Remark 3.1 let $\mathbb{E}[X_i] = 0$.

Step 1 (the subsequence $n_k = k^2$). For $\varepsilon > 0$, Chebyshev gives $P(|S_{k^2}|/k^2 > \varepsilon) \le \frac{\mathrm{Var}(S_{k^2})}{\varepsilon^2 k^4} \le \frac{c}{\varepsilon^2 k^2}$, which is summable. By Lemma 2.10, $S_{k^2}/k^2 \to 0$ outside a null set N_1.

Step 2 (fluctuations between squares). Let $D_k := \max_{k^2 \le l < (k+1)^2} |S_l - S_{k^2}|$. Since $\mathrm{Var}(S_l - S_{k^2}) = \sum_{k^2 < i \le l} \sigma_i^2 \le 2kc$ and there are $2k$ indices l,

$$P\Big(\frac{D_k}{k^2} \ge \varepsilon\Big) \le \sum_{l=k^2+1}^{k^2+2k} P(|S_l - S_{k^2}| \ge \varepsilon k^2) \le 2k \cdot \frac{2kc}{\varepsilon^2 k^4} = \frac{4c}{\varepsilon^2 k^2}.$$

Lemma 2.10 gives $D_k/k^2 \to 0$ outside a null set N_2.

Step 3. For $k^2 \le n < (k+1)^2$: $\frac{|S_n|}{n} \le \frac{|S_{k^2}| + D_k}{k^2} \to 0$ on $(N_1 \cup N_2)^c$. ∎

Example 3.8

For Bernoulli experiments, $\frac{1}{n}\sum_{i \le n} X_i \to p$ P_p-a.s. (É. Borel 1909, strengthening Bernoulli's (3.1)). For a fair coin, $Y_i := 2X_i - 1 \in \{\pm 1\}$ and $S_n = Y_1 + \cdots + Y_n$ is the position of a **simple random walk** on $\mathbb{Z}$; refined, rescaled and linearly interpolated it converges to Brownian motion (Section 4.1). The strong law gives $S_n/n \to 0$ a.s.: fluctuations grow slower than linearly. Their exact order is given by the *law of the iterated logarithm*:

$$\limsup_n \frac{S_n}{\sqrt{2n \log\log n}} = +1, \qquad \liminf_n \frac{S_n}{\sqrt{2n \log\log n}} = -1 \qquad P\text{-a.s.}$$

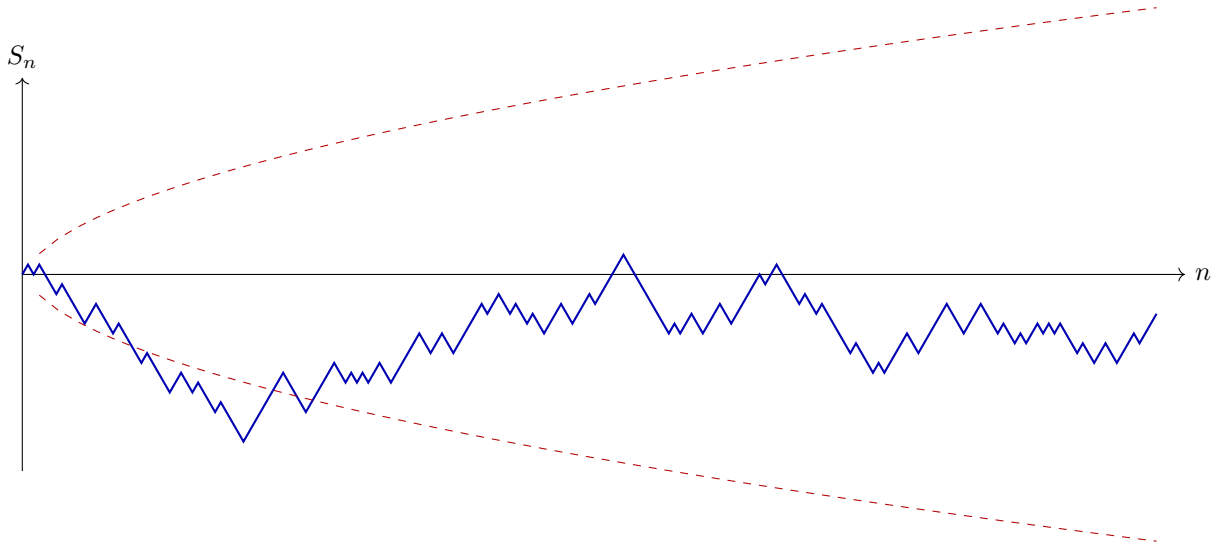


Figure. *A simple random walk path (200 steps). It is much smaller than n (strong law) but keeps returning to the order of $\sqrt{n \log\log n}$ (dashed: $\pm\sqrt{2n \log\log n}$, the scale in the law of the iterated logarithm).*

3.4 Independence

Motivation. Independence is the structural assumption behind most limit theorems. It is defined through products of probabilities on finitely many events at a time. With the π–λ theorem it passes from generators to σ-algebras, which gives Kolmogorov's 0–1 law and the second Borel–Cantelli lemma.

3.4.1 Independent events

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

Definition 3.9 (Independence)

Events $A_i \in \mathcal{A}$, $i \in I$, are **independent** if $P(\bigcap_{j\in J} A_j) = \prod_{j\in J} P(A_j)$ for every finite $J \subset I$. Families $\mathcal{B}_i \subset \mathcal{A}$, $i \in I$, are **independent** if this holds for all finite $J \subset I$ and all choices $A_j \in \mathcal{B}_j$.

Theorem 3.10

Let $\mathcal{B}_i$, $i \in I$, be independent $\cap$-stable families. Then (i) $\sigma(\mathcal{B}_i)$, $i \in I$, are independent; (ii) for a partition $I = \bigcup_{k\in K} J_k$, the σ-algebras $\sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$, $k \in K$, are independent.

Proof. (i) Let $J = \{j_1, \dots, j_n\}$ and $A_{j_\nu} \in \sigma(\mathcal{B}_{j_\nu})$; we show

$$P(A_{j_1} \cap \cdots \cap A_{j_n}) = P(A_{j_1}) \cdots P(A_{j_n}). \tag{3.2}$$

Step 1. Fix $A_{j_2} \in \mathcal{B}_{j_2}, \dots, A_{j_n} \in \mathcal{B}_{j_n}$. Then $\mathcal{D}_{j_1} := \{A \in \sigma(\mathcal{B}_{j_1}) \mid P(A \cap A_{j_2} \cap \cdots \cap A_{j_n}) = P(A)P(A_{j_2}) \cdots P(A_{j_n})\}$ is a Dynkin system containing $\mathcal{B}_{j_1}$, so $\mathcal{D}_{j_1} = \sigma(\mathcal{B}_{j_1})$ by Theorem 1.8.

Step 2. Now $\sigma(\mathcal{B}_{j_1}), \mathcal{B}_{j_2}, \dots, \mathcal{B}_{j_n}$ are independent $\cap$-stable families; repeat Step 1 for j_2, then j_3, and so on, to get (3.2).

(ii) For $k \in K$ let $\mathcal{C}_k := \{\bigcap_{j\in J} A_j \mid J \subset J_k \text{ finite}, \ A_j \in \mathcal{B}_j\}$. Each $\mathcal{C}_k$ is $\cap$-stable, and the $\mathcal{C}_k$ are independent: for $k_1, \dots, k_n$ and finite $J_\nu \subset J_{k_\nu}$, independence of the $\mathcal{B}_i$ gives $P(\bigcap_\nu \bigcap_{i\in J_\nu} A_i) = \prod_\nu P(\bigcap_{i\in J_\nu} A_i)$. By (i), the $\sigma(\mathcal{C}_k)$ are independent. Finally $\bigcup_{i\in J_k} \mathcal{B}_i \subset \mathcal{C}_k \subset \sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$, so $\sigma(\mathcal{C}_k) = \sigma(\bigcup_{i\in J_k} \mathcal{B}_i)$. ∎

Example 3.11

If A_i, $i \in I$, are independent, then the σ-algebras $\sigma(\{A_i\}) = \{\emptyset, A_i, A_i^c, \Omega\}$ are independent (Theorem 3.10(i) with $\mathcal{B}_i = \{A_i\}$). The collection of events $\{A_i, A_i^c \mid i \in I\}$, however, is in general not independent: $P(A_i \cap A_i^c) = 0 \neq P(A_i)P(A_i^c)$ unless $P(A_i) \in \{0, 1\}$.

Remark 3.12

Pairwise independence does not imply independence. Toss a fair coin twice, $\Omega = \{0,1\}^2$ uniform, and let $A =$ "first toss 1", $B =$ "second toss 1", $C =$ "both tosses equal". All have probability $\frac{1}{2}$ and every pairwise intersection has probability $\frac{1}{4}$, so they are pairwise independent; but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C)$.

Example 3.13 (Independent 0-1 experiments)

On $\Omega = \{0,1\}^{\mathbb{N}}$ with $\mathcal{A} = \sigma(\{X_i = 1\}, i \in \mathbb{N})$ let P_p satisfy (i) $P_p(X_i = 1) = p$ and (ii) the events $\{X_i = 1\}$, $i \in \mathbb{N}$, are independent. By (ii) and Example 3.11, for $i_1 < \cdots < i_n$ and $x_j \in \{0,1\}$

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = \prod_j P_p(X_{i_j} = x_j) = p^k(1-p)^{n-k}, \qquad k = \textstyle\sum_j x_j.$$

These cylinder probabilities form a $\cap$-stable generator, so P_p is uniquely determined by (i) and (ii) (Theorem 1.9).

Theorem 3.14 (Kolmogorov's 0–1 law)

Let $\mathcal{B}_n$, $n \in \mathbb{N}$, be independent σ-algebras and $\mathcal{B}_\infty := \bigcap_{n=1}^\infty \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$ the **tail σ-algebra**. Then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}_\infty$.

Illustration. For 0-1 experiments with $\mathcal{B}_i = \sigma(\{X_i = 1\})$, $\mathcal{B}_\infty$ contains the events of the remote future, e.g. $\limsup_i \{X_i = 1\}$ ("infinitely many 1's") and $\{\lim_n \frac{1}{n} \sum_{i \leq n} X_i \text{ exists}\}$.

Proof.

Step 1. By Theorem 3.10(ii), for every $n \geq 2$ the σ-algebras $\mathcal{B}_1, \dots, \mathcal{B}_{n-1}, \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$ are independent. Since $\mathcal{B}_\infty \subset \sigma(\bigcup_{m \geq n} \mathcal{B}_m)$, also $\mathcal{B}_1, \dots, \mathcal{B}_{n-1}, \mathcal{B}_\infty$ are independent for all n, i.e. $\mathcal{B}_\infty, \mathcal{B}_1, \mathcal{B}_2, \dots$ are independent.

Step 2. Theorem 3.10(ii) again: $\sigma(\bigcup_n \mathcal{B}_n)$ and $\mathcal{B}_\infty$ are independent. As $\mathcal{B}_\infty \subset \sigma(\bigcup_n \mathcal{B}_n)$, $\mathcal{B}_\infty$ is independent of itself; conclude with Lemma 3.15. ∎

Lemma 3.15

If a σ-algebra $\mathcal{B} \subset \mathcal{A}$ is independent of itself, then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}$.

Proof. $P(A) = P(A \cap A) = P(A)^2$. ∎

In particular, for independent events A_n, $A_\infty := \limsup_n A_n$ has $P(A_\infty) \in \{0,1\}$: the σ-algebras $\mathcal{B}_n := \sigma(\{A_n\})$ are independent (Example 3.11) and $A_\infty \in \mathcal{B}_\infty$. Which value occurs is decided by:

Lemma 3.16 (Borel–Cantelli, part 2)

If $A_i \in \mathcal{A}$ are independent and $\sum_i P(A_i) = \infty$, then $P(\limsup_i A_i) = 1$.

Proof. It suffices that $P(\bigcap_{m \geq n} A_m^c) = 0$ for every n. By continuity from above, independence of the complements (Example 3.11) and $1 - \alpha \leq e^{-\alpha}$,

$$P\Big(\bigcap_{m \geq n} A_m^c\Big) = \lim_k P\Big(\bigcap_{m=n}^{n+k} A_m^c\Big) = \lim_k \prod_{m=n}^{n+k} (1 - P(A_m)) \leq \lim_k \exp\Big(-\sum_{m=n}^{n+k} P(A_m)\Big) = 0.$$

∎

Example 3.17 (A text appears infinitely often)

Let $p \in (0,1)$ and fix a binary word $(x_1, \dots, x_N)$ with K ones. Cut $\omega = (y_n)$ into consecutive blocks of length N and let A_i = "the word occurs as the i-th block". By Theorem 3.10(ii) the A_i are independent, each with probability $\alpha := p^K(1-p)^{N-K} > 0$. So $\sum_i P_p(A_i) = \infty$

and Borel–Cantelli 2 gives $P_p(\limsup_i A_i) = 1$: the word occurs in infinitely many blocks. Moreover the $\mathbb{1}_{A_i}$ are independent, hence uncorrelated, with bounded variances, so the strong law gives $\frac{1}{n}\sum_{i\le n}\mathbb{1}_{A_i} \to \alpha > 0$ P_p-a.s.: the word has a strictly positive relative frequency.

3.4.2 Independent random variables

Definition 3.18 (Independent random variables)

(i) For a r.v. $X : (\Omega, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $\sigma(X) := X^{-1}(\mathcal{B}(\mathbb{R})) = \{\{X \in A\} \mid A \in \mathcal{B}(\mathbb{R})\} \subset \mathcal{A}$ is the σ-algebra **generated by** X.

(ii) R.v.'s X_i, $i \in I$, are **independent** if the $\sigma(X_i)$ are independent, i.e. $P(\bigcap_{j\in J}\{X_j \in A_j\}) = \prod_{j\in J} P(X_j \in A_j)$ for finite J and Borel A_j.

(iii) X_i, $i \in I$, are **independent of** a sub-σ-algebra $\tilde{\mathcal{A}}$ if $\sigma(X_i \mid i \in I)$ (the smallest σ-algebra making all X_i measurable) is independent of $\tilde{\mathcal{A}}$.

Remark 3.19

If X_i, $i \in I$, are independent and $h_i : \mathbb{R} \to \mathbb{R}$ are Borel, then $Y_i := h_i(X_i)$ are independent, because $\sigma(Y_i) \subset \sigma(X_i)$.

Theorem 3.20 (Product rule)

If $X_1, \dots, X_n \ge 0$ are independent, then $\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n]$.

Proof.

Step 1 (reduction to $n = 2$). $X_1 \cdots X_{n-1}$ is measurable w.r.t. $\sigma(\sigma(X_1) \cup \cdots \cup \sigma(X_{n-1}))$, which is independent of $\sigma(X_n)$ by Theorem 3.10(ii); induct. So let $X, Y \ge 0$ be independent; we show

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]. \tag{3.3}$$

Step 2 (simple functions). For $X_n = \sum_i \alpha_i \mathbb{1}_{A_i}$, $Y_n = \sum_j \beta_j \mathbb{1}_{B_j}$ with $\alpha_i, \beta_j \ge 0$, $A_i \in \sigma(X)$, $B_j \in \sigma(Y)$:

$$\mathbb{E}[X_n Y_n] = \sum_{i,j} \alpha_i \beta_j P(A_i \cap B_j) = \sum_{i,j} \alpha_i \beta_j P(A_i) P(B_j) = \mathbb{E}[X_n]\mathbb{E}[Y_n].$$

Step 3. Choose $\sigma(X)$- resp. $\sigma(Y)$-measurable simple $X_n \nearrow X$, $Y_n \nearrow Y$ (Facts 2.2(v)); then $X_n Y_n \nearrow XY$, and the MCT turns Step 2 into (3.3). ∎

Corollary 3.21

X, Y independent and in $\mathcal{L}^1 \Rightarrow XY \in \mathcal{L}^1$ and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. For $\varepsilon_1, \varepsilon_2 \in \{+, -\}$, X^{ε_1} and Y^{ε_2} are independent (Remark 3.19) and nonnegative, so $\mathbb{E}[X^{\varepsilon_1} Y^{\varepsilon_2}] = \mathbb{E}[X^{\varepsilon_1}]\mathbb{E}[Y^{\varepsilon_2}] < \infty$. Hence $XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+ \in \mathcal{L}^1$ and expanding gives $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. ∎

Remark 3.22

The converse fails. For $X \sim N(0,1)$ and $Y = X^2$, X, Y are dependent but $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$. Uncorrelatedness is also not preserved under functions: $\mathbb{E}[X^+Y^+] \neq \mathbb{E}[X^+]\mathbb{E}[Y^+]$. [Supplement] Indeed $\mathbb{E}[X^+Y^+] = \int_0^\infty x^3\varphi(x)\,dx = \frac{2}{\sqrt{2\pi}}$ (since $\int_0^\infty x^3 e^{-x^2/2}dx = 2$), while $\mathbb{E}[X^+]\mathbb{E}[Y^+] = \frac{1}{\sqrt{2\pi}} \cdot 1$.
Obviously $X, Y \in \mathcal{L}^2$ independent $\Rightarrow$ uncorrelated, since $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$.

Corollary 3.23 (Strong law for independent variables)

If $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\sup_i \mathrm{Var}(X_i) < \infty$, then $\frac{1}{n}\sum_{i\le n}(X_i - \mathbb{E}[X_i]) \to 0$ a.s.; if $\mathbb{E}[X_i] \equiv m$, then $\frac{1}{n}\sum_{i\le n} X_i \to m$ a.s.

Proof. Independent $\mathcal{L}^2$ variables are uncorrelated (Remark 3.22); apply Theorem 3.7. ∎

3.5 Kolmogorov's law of large numbers

Motivation. The strong law of Section 3.3 needs second moments. For identically distributed variables a first moment suffices, and pairwise independence is enough. The proof truncates (Borel–Cantelli removes the truncation), uses Chebyshev along geometric subsequences, and interpolates by monotonicity.

(X_i) is **identically distributed** (i.d.) if $P \circ X_i^{-1} = P \circ X_1^{-1}$ for all i; i.i.d. means independent and identically distributed.

Theorem 3.24 (Kolmogorov 1930)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are i.i.d. with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. Independent variables are in particular pairwise independent; apply Theorem 3.25. ∎

It follows from the more general:

Theorem 3.25 (Etemadi 1981)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are pairwise independent and identically distributed with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. We may assume $X_i \ge 0$; otherwise treat (X_i^+) and (X_i^-), which are again in $\mathcal{L}^1$, pairwise independent and identically distributed.

Step 1 (truncation). Let $\tilde{X}_i := X_i \mathbb{1}_{\{X_i < i\}} = h_i(X_i)$ with $h_i(x) = x\mathbb{1}_{\{x<i\}}$; the $\tilde{X}_i$ are pairwise independent (Remark 3.19). It suffices to show $\tilde{S}_n/n \to m$ a.s. for $\tilde{S}_n := \sum_{i\le n}\tilde{X}_i$, because

$$\sum_n P(X_n \neq \tilde{X}_n) = \sum_n P(X_1 \ge n) = \sum_n \sum_{k\ge n} P(X_1 \in [k, k+1))$$
$$= \sum_k k\, P(X_1 \in [k, k+1)) \le \mathbb{E}[X_1] < \infty,$$

so by Borel–Cantelli $X_n = \tilde{X}_n$ for all but finitely many n, a.s.

Step 2 (geometric subsequence). Fix $\alpha > 1$ and $k_n := \lfloor \alpha^n \rfloor$. Suppose (Step 3)

$$\frac{\tilde{S}_{k_n} - \mathbb{E}[\tilde{S}_{k_n}]}{k_n} \to 0 \quad \text{a.s.} \tag{3.4}$$

Since $\mathbb{E}[\tilde{X}_i] = \mathbb{E}[X_1 \mathbb{1}_{\{X_1<i\}}] \nearrow m$, the Cesàro means $\frac{1}{k_n}\mathbb{E}[\tilde{S}_{k_n}] \to m$, so $\tilde{S}_{k_n}/k_n \to m$ outside a null set N_α. For $k_n \le l < k_{n+1}$, monotonicity of $l \mapsto \tilde{S}_l$ gives

$$\frac{k_n}{k_{n+1}} \cdot \frac{\tilde{S}_{k_n}}{k_n} \le \frac{\tilde{S}_l}{l} \le \frac{\tilde{S}_{k_{n+1}}}{k_{n+1}} \cdot \frac{k_{n+1}}{k_n},$$

and $k_{n+1}/k_n \to \alpha$. Hence $\frac{m}{\alpha} \le \liminf_l \frac{\tilde{S}_l}{l} \le \limsup_l \frac{\tilde{S}_l}{l} \le \alpha m$ off N_α. Taking $\alpha_j \searrow 1$ and the null set $\bigcup_j N_{\alpha_j}$ gives $\tilde{S}_l/l \to m$ a.s.

Step 3 (proof of (3.4)). By Lemma 2.10 it suffices to show $\sum_n P(|\tilde{S}_{k_n} - \mathbb{E}[\tilde{S}_{k_n}]| \ge \varepsilon k_n) < \infty$. Pairwise independent $\Rightarrow$ pairwise uncorrelated, so Chebyshev gives the bound $\frac{1}{\varepsilon^2 k_n^2}\sum_{i\le k_n}\mathbb{E}[\tilde{X}_i^2]$, and it is enough that

$$s := \sum_n \frac{1}{k_n^2}\sum_{i\le k_n}\mathbb{E}[\tilde{X}_i^2] = \sum_i \Big(\sum_{n:\, k_n \ge i}\frac{1}{k_n^2}\Big)\mathbb{E}[\tilde{X}_i^2] < \infty.$$

Step 4 (a geometric bound). We claim there is c with

$$\sum_{n:\, k_n \ge i}\frac{1}{k_n^2} \le \frac{c}{i^2}. \tag{3.5}$$

Indeed $k_n > \alpha^n - 1 \ge \alpha^n - \alpha^{n-1} = c_\alpha \alpha^n$ with $c_\alpha = \frac{\alpha-1}{\alpha}$. If n_i is the least n with $k_n \ge i$, then $\alpha^{n_i} \ge i$ and

$$\sum_{n\ge n_i}\frac{1}{k_n^2} \le c_\alpha^{-2}\sum_{n\ge n_i}\alpha^{-2n} = \frac{c_\alpha^{-2}\alpha^{-2n_i}}{1-\alpha^{-2}} \le \frac{c_\alpha^{-2}}{1-\alpha^{-2}}\cdot\frac{1}{i^2}.$$

Step 5. With (3.5), $\mathbb{E}[\tilde{X}_i^2] = \mathbb{E}[X_1^2\mathbb{1}_{\{X_1<i\}}] \le \sum_{l\le i} l^2 P(X_1 \in [l-1,l))$ and $\sum_{i\ge l}\frac{1}{i^2} \le \frac{1}{l^2} + \sum_{i>l}\left(\frac{1}{i-1} - \frac{1}{i}\right) \le \frac{2}{l}$:

$$\begin{aligned} s &\le c\sum_i \frac{1}{i^2}\sum_{l\le i} l^2 P(X_1 \in [l-1,l)) = c\sum_l l^2\Big(\sum_{i\ge l}\frac{1}{i^2}\Big)P(X_1 \in [l-1,l)) \\ &\le 2c\sum_l \mathbb{E}\big[l\mathbb{1}_{\{X_1\in[l-1,l)\}}\big] \le 2c(\mathbb{E}[X_1]+1). \end{aligned}$$

■

Corollary 3.26

If $X_1, X_2, \cdots \ge 0$ are pairwise independent and identically distributed, then $\frac{1}{n}\sum_{i\le n} X_i \to \mathbb{E}[X_1] \in [0,\infty]$ a.s.

Proof. If $\mathbb{E}[X_1] < \infty$ this is Theorem 3.25. If $\mathbb{E}[X_1] = \infty$, apply Theorem 3.25 to $X_i \wedge N$:

$$\frac{1}{n}\sum_{i\le n} X_i \ge \frac{1}{n}\sum_{i\le n}(X_i \wedge N) \to \mathbb{E}[X_1 \wedge N] \quad \text{a.s.},$$

and $\mathbb{E}[X_1 \wedge N] \nearrow \mathbb{E}[X_1] = \infty$ as $N \to \infty$ (MCT). Hence $\liminf_n \frac{1}{n}\sum_{i\le n} X_i = \infty$ a.s. ■

3.6 Finite products of probability measures

Motivation. Independent experiments are modelled on a product space whose measure multiplies the probabilities of rectangles. We construct it by integrating sections, which gives Fubini's theorem at the same time: integrals over a product can be computed as iterated integrals in any order.

Let $(S_1, \mathcal{S}_1, \mu_1)$, $(S_2, \mathcal{S}_2, \mu_2)$ be probability spaces, $\Omega := S_1 \times S_2$, $X_i(x_1, x_2) = x_i$, and

$$\mathcal{A} := \mathcal{S}_1 \otimes \mathcal{S}_2 := \sigma(X_1, X_2) = \sigma(\{A_1 \times A_2 \mid A_i \in \mathcal{S}_i\}).$$

We seek a probability measure $P =: \mu_1 \otimes \mu_2$ (the **product measure**) with

$$P(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \qquad (A_i \in \mathcal{S}_i). \tag{3.6}$$

Such P is unique, since the rectangles $A_1 \times A_2$ form a $\cap$-stable generator of $\mathcal{A}$ (Theorem 1.9).

Theorem 3.27 (Fubini for probability measures)

(i) For every $\mathcal{A}$-measurable $f : \Omega \to \overline{\mathbb{R}}$ and $x_1 \in S_1$, $x_2 \in S_2$, the sections $f_{x_1} := f(x_1, \cdot)$ and $f_{x_2} := f(\cdot, x_2)$ are measurable. If moreover $f \geq 0$, then $x_1 \mapsto \int_{S_2} f_{x_1}\, d\mu_2$ and $x_2 \mapsto \int_{S_1} f_{x_2}\, d\mu_1$ are $\mathcal{S}_1$- resp. $\mathcal{S}_2$-measurable.

(ii) There is a unique probability measure $P = \mu_1 \otimes \mu_2$ on $(\Omega, \mathcal{A})$ such that for all $\mathcal{A}$-measurable $f \geq 0$

$$\int_\Omega f\, dP = \int_{S_1} \Big(\int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1), \tag{3.7}$$

and the order may be interchanged:

$$\int_\Omega f\, dP = \int_{S_2} \Big(\int_{S_1} f(x_1, x_2)\, \mu_1(dx_1) \Big) \mu_2(dx_2). \tag{3.8}$$

In particular, with the sections $A_{x_1} := \{x_2 \mid (x_1, x_2) \in A\}$, $A_{x_2} := \{x_1 \mid (x_1, x_2) \in A\}$,

$$P(A) = \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1) = \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2) \qquad (A \in \mathcal{A}). \tag{3.9}$$

Proof.

Step 1 (sections are measurable). $\phi_{x_1}(x_2) := (x_1, x_2)$ is measurable, because $\phi_{x_1}^{-1}(A_1 \times A_2)$ is A_2 or $\emptyset$ and rectangles generate $\mathcal{A}$ (Remark 1.32). So $f_{x_1} = f \circ \phi_{x_1}$ is measurable; likewise f_{x_2}.

Step 2 (indicators). For $f \geq 0$, the map

$$x_1 \mapsto \int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \in [0, \infty] \tag{3.10}$$

is well defined. For $f = \mathbb{1}_A$ it equals $\mu_2(A_{x_1})$. Let $\mathcal{D} := \{A \in \mathcal{A} \mid x_1 \mapsto \mu_2(A_{x_1}) \text{ is } \mathcal{S}_1\text{-measurable}\}$. $\mathcal{D}$ is a Dynkin system: $\Omega_{x_1} = S_2$; $\mu_2((A^c)_{x_1}) = 1 - \mu_2(A_{x_1})$; for disjoint A_n the sections are disjoint and $\mu_2((\bigcup A_n)_{x_1}) = \sum_n \mu_2((A_n)_{x_1})$ is a series of measurable functions. $\mathcal{D}$ contains the rectangles, since $\mu_2((A_1 \times A_2)_{x_1}) = \mathbb{1}_{A_1}(x_1)\mu_2(A_2)$. By Theorem 1.8, $\mathcal{D} = \mathcal{A}$.

Step 3 (measure-theoretic induction). By linearity (3.10) is measurable for simple $f \geq 0$; for general $f \geq 0$ take simple $f_n \nearrow f$ (Facts 2.2(v)). The MCT makes (3.10) a pointwise limit of measurable functions. The other order is identical.

Step 4 (existence). Define $P(A) := \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1)$. Then $P(\Omega) = 1$, and for disjoint A_n the MCT (Theorem 1.56) gives $P(\bigcup_n A_n) = \int \sum_n \mu_2((A_n)_{x_1})\, d\mu_1 = \sum_n P(A_n)$. Also $P(A_1 \times A_2) = \int \mathbb{1}_{A_1} \mu_2(A_2)\, d\mu_1 = \mu_1(A_1)\mu_2(A_2)$, so (3.6) holds and P is unique. (3.7) holds for indicators by definition and extends to $f \geq 0$ by linearity and the MCT.

Step 5 (interchanging the order). Repeating Steps 2–4 with the coordinates swapped, $Q(A) := \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2)$ is a probability measure with $Q(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$, so $Q = P$ by Theorem 1.9. Hence (3.8) holds for indicators and, by linearity and the MCT, for all $f \geq 0$; (3.9) is the case $f = \mathbb{1}_A$. ∎

Remark 3.28

If $f \in \mathcal{L}^1(\mu_1 \otimes \mu_2)$, Theorem 3.27 applies to f^+ and f^-, hence by linearity to f.

Definition 3.29 (Finite products of measurable spaces)

For measurable spaces $(S_i, \mathcal{S}_i)$, $1 \le i \le n$, $\Omega := S_1 \times \cdots \times S_n$ carries the **product σ-algebra**

$$\mathcal{A} := \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n = \bigotimes_{i=1}^{n} \mathcal{S}_i := \sigma(\{A_1 \times \cdots \times A_n \mid A_i \in \mathcal{S}_i\}).$$

Equivalently $\mathcal{A} = \sigma(X_1, \dots, X_n)$ with the coordinate projections X_i. [Supplement] *Reason:* $A_1 \times \cdots \times A_n = \bigcap_i X_i^{-1}(A_i)$, and conversely $X_i^{-1}(A_i)$ is the rectangle with A_i in slot i and S_j elsewhere.

Theorem 3.30 (Finite products of probability measures)

For probability spaces $(S_i, \mathcal{S}_i, \mu_i)$, $1 \le i \le n$, there is a unique probability measure $P = \mu_1 \otimes \cdots \otimes \mu_n = \bigotimes_{i=1}^n \mu_i$ on $(S_1 \times \cdots \times S_n, \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n)$ with $P(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$.

Proof. Uniqueness: rectangles form a $\cap$-stable generator (Theorem 1.9). Existence by induction, $n = 2$ being Theorem 3.27. Given $P_{n-1} := \mu_1 \otimes \cdots \otimes \mu_{n-1}$ on $(\Omega_{n-1}, \mathcal{A}_{n-1})$, apply Theorem 3.27 to $(\Omega_{n-1}, \mathcal{A}_{n-1}, P_{n-1})$ and $(S_n, \mathcal{S}_n, \mu_n)$. Since $\Omega_{n-1} \times S_n = S_1 \times \cdots \times S_n$ and $\mathcal{A}_{n-1} \otimes \mathcal{S}_n = \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$ (both are generated by the n-fold rectangles), $P := P_{n-1} \otimes \mu_n$ lives on the right space, and $P(A_1 \times \cdots \times A_n) = P_{n-1}(A_1 \times \cdots \times A_{n-1})\mu_n(A_n) = \prod_i \mu_i(A_i)$. ∎

Theorem 3.31 (Fubini for finite products)

With $P = \mu_1 \otimes \cdots \otimes \mu_n$ and $\mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$-measurable $f \ge 0$,

$$\int f \, dP = \int_{S_1} \cdots \int_{S_n} f(x_1, \dots, x_n) \, \mu_n(dx_n) \cdots \mu_1(dx_1), \tag{3.11}$$

and for every permutation π of $\{1, \dots, n\}$

$$\int f \, dP = \int_{S_{\pi(1)}} \cdots \int_{S_{\pi(n)}} f(x_1, \dots, x_n) \, \mu_{\pi(n)}(dx_{\pi(n)}) \cdots \mu_{\pi(1)}(dx_{\pi(1)}). \tag{3.12}$$

Proof.

Step 1 ((3.11)). Induction on n; $n = 2$ is Theorem 3.27. With $P = P_{n-1} \otimes \mu_n$ (proof of Theorem 3.30), the two-factor Fubini theorem gives $\int f \, dP = \int_{\Omega_{n-1}} \left(\int_{S_n} f \, d\mu_n \right) dP_{n-1}$. The inner integral is a nonnegative $\mathcal{A}_{n-1}$-measurable function, to which the induction hypothesis applies.

Step 2 ((3.12)) [Supplement]. Regrouping Ω as $\left(\prod_{i \ne \pi(n)} S_i\right) \times S_{\pi(n)}$ identifies the product σ-algebras (both are generated by rectangles). Hence, by Theorem 3.27(i) and induction, all iterated integrals in (3.12) are well defined and measurable at each stage. For $f = \mathbb{1}_A$ the right-hand side of (3.12), as a function of A, is a probability measure (MCT for disjoint unions) which gives rectangles the value $\prod_i \mu_i(A_i)$. By Theorem 1.9 it equals P. Linearity and the MCT extend this to all $f \ge 0$. ∎

Remark 3.32

If $f \in \mathcal{L}^1(\mu_1 \otimes \cdots \otimes \mu_n)$, Theorem 3.31 applies to $f^\pm$ and hence to f.

3.7 Weak convergence of probability measures

Motivation. Laws of averages such as the central limit theorem compare distributions, not random variables on one space. Requiring $\mu_n(A) \to \mu(A)$ for every set is far too strong ($\delta_{1/n} \not\to \delta_0$ on $\{0\}$). Weak convergence tests against bounded continuous functions, and the Portmanteau theorem lists equivalent, more concrete forms.

Definition 3.33 (Weak convergence)

Let Ω be a metric space with Borel σ-algebra $\mathcal{B}(\Omega)$, and μ, μ_n probability measures on it. $\mu_n \to \mu$ **weakly** if $\int f\,d\mu_n \to \int f\,d\mu$ for all $f \in C_b(\Omega)$ (bounded continuous functions).

Example 3.34

(i) $x_n \to x$ implies $\delta_{x_n} \to \delta_x$ weakly, since $f(x_n) \to f(x)$ for continuous f. (ii) $\mu_n := N(0, \frac{1}{n}) \to \delta_0$ weakly: substituting $x = y/\sqrt{n}$ and using the DCT,

$$\int f\,d\mu_n = \int f(x)\frac{e^{-nx^2/2}}{\sqrt{2\pi/n}}\,dx = \int f\Big(\frac{y}{\sqrt{n}}\Big)\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy \to f(0) = \int f\,d\delta_0.$$

The following results also hold for finite measures with the obvious changes.

Theorem 3.35 (Portmanteau theorem)

For probability measures μ, μ_n on a metric space (Ω, d) the following are equivalent:

(i) $\mu_n \to \mu$ weakly;

(ii) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded uniformly continuous f;

(iii) $\limsup_n \mu_n(F) \le \mu(F)$ for all closed F;

(iv) $\liminf_n \mu_n(G) \ge \mu(G)$ for all open G;

(v) $\mu_n(A) \to \mu(A)$ for all μ-continuity sets A, i.e. $A \in \mathcal{B}(\Omega)$ with $\mu(\partial A) = 0$;

(vi) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded measurable f that are continuous μ-a.e.

Proof.

Step Easy implications. (i)⇒(ii) and (vi)⇒(i) are special cases; (iii)⇔(iv) by taking complements.

Step (ii)⇒(iii). For closed F the open sets $G_m := \{d(\cdot, F) < \frac{1}{m}\}$ decrease to F, so given ε there is m with $\mu(G_m) < \mu(F) + \varepsilon$. Let $\phi(x) = 1$ for $x \le 0$, $1 - x$ on $[0,1]$, 0 for $x \ge 1$, and $f := \phi(m\,d(\cdot, F))$; it is uniformly continuous (composition of Lipschitz maps) with $\mathbb{1}_F \le f \le \mathbb{1}_{G_m}$. Hence $\limsup_n \mu_n(F) \le \lim_n \int f\,d\mu_n = \int f\,d\mu \le \mu(G_m) < \mu(F) + \varepsilon$.

Step (iii)⇒(v). With interior $\mathring{A}$ and closure $\bar{A}$, $\mu(\partial A) = 0$ gives $\mu(\mathring{A}) = \mu(A) = \mu(\bar{A})$, and

$$\mu(A) = \mu(\mathring{A}) \overset{\text{(iv)}}{\le} \liminf_n \mu_n(\mathring{A}) \le \liminf_n \mu_n(A) \le \limsup_n \mu_n(A) \le \limsup_n \mu_n(\bar{A}) \overset{\text{(iii)}}{\le} \mu(\bar{A}) = \mu(A).$$

Step (v)⇒(vi). Let f be as in (vi) and $D := \{x \in \mathbb{R} \mid \mu(f = x) \ne 0\}$, the jumps of the distribution function of $\mu \circ f^{-1}$; it is countable (Remark 2.16), so $\mathbb{R} \setminus D$ is dense. Given $\varepsilon > 0$ pick $c_0 < \cdots < c_m$ in $\mathbb{R} \setminus D$ with $c_{k+1} - c_k \le \varepsilon$ and $c_0 \le f < c_m$, and set $A_k := f^{-1}([c_k, c_{k+1}))$. Each A_k is a μ-continuity set, because (see below)

$$\partial A_k \subset \{f = c_k\} \cup \{f = c_{k+1}\} \cup D_f, \qquad D_f := \{\omega \mid f \text{ is discontinuous at } \omega\}, \tag{3.13}$$

and all three sets are μ-null. With $g := \sum_k c_k \mathbb{1}_{A_k}$, $\|f-g\|_\infty \le \varepsilon$ and

$$\Big|\int f\,d\mu - \int f\,d\mu_n\Big| \le \int |f-g|\,d\mu + \Big|\int g\,d\mu - \int g\,d\mu_n\Big| + \int |g-f|\,d\mu_n$$
$$\le 2\varepsilon + \sum_k |c_k||\mu(A_k) - \mu_n(A_k)| \to 2\varepsilon.$$

Step Proof of (3.13). Let $\omega \in \partial A_k \setminus D_f$. For every $\varepsilon' > 0$ there is δ with $f(B(\omega,\delta)) \subset B(f(\omega),\varepsilon')$, and $B(\omega,\delta)$ meets both $f^{-1}([c_k,c_{k+1}))$ and its complement. Applying f, $B(f(\omega),\varepsilon')$ meets both $[c_k,c_{k+1})$ and its complement, for every ε'. So $f(\omega) \in \partial[c_k,c_{k+1}) = \{c_k, c_{k+1}\}$.

[Supplement] D_f is measurable: with the oscillation $\operatorname{osc}_f(\omega) := \lim_{\delta\to 0} \sup\{|f(x)-f(y)| \mid x,y \in B(\omega,\delta)\}$, $D_f = \bigcup_k \{\operatorname{osc}_f \ge \frac{1}{k}\}$, and each $\{\operatorname{osc}_f \ge \frac{1}{k}\}$ is closed (if $\operatorname{osc}_f(\omega) < \frac{1}{k}$, the same small ball works for nearby points). ∎

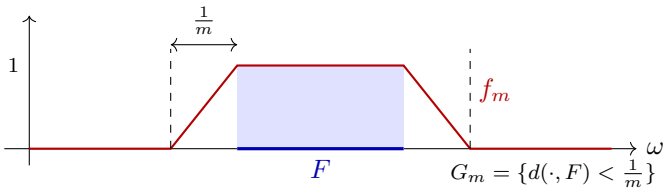


Figure. *The test function in (ii)⇒(iii): $f_m = \phi(m\,d(\cdot,F))$ is 1 on F, 0 off the $\frac{1}{m}$-neighbourhood G_m, and linear in between, so $\mathbb{1}_F \le f_m \le \mathbb{1}_{G_m} \searrow \mathbb{1}_F$.*

Definition 3.36 (Support, C_0)

For continuous f on a metric space Ω, $\operatorname{supp} f := \overline{\{f \neq 0\}}$, and $C_0(\Omega) := \{f \in C_b(\Omega) \mid \operatorname{supp} f \text{ compact}\}$.

Corollary 3.37

For probability measures μ, μ_n on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with distribution functions F, F_n, the following are equivalent:

(i) $\mu_n \to \mu$ **vaguely**: $\int f\,d\mu_n \to \int f\,d\mu$ for all $f \in C_0(\mathbb{R})$;

(ii) $\mu_n \to \mu$ weakly;

(iii) $F_n(x) \to F(x)$ at every continuity point x of F;

(iv) $\mu_n((a,b]) \to \mu((a,b])$ whenever $\mu(\{a\}) = \mu(\{b\}) = 0$.

Proof.

Step (i)⇒(ii) [Supplement]. Let $f \in C_b(\mathbb{R})$, $\varepsilon > 0$. Choose R with $\mu([-R,R]) > 1-\varepsilon$ and $g \in C_0(\mathbb{R})$, $0 \le g \le 1$, $g = 1$ on $[-R,R]$. Then $\int g\,d\mu_n \to \int g\,d\mu > 1-\varepsilon$, so $\int (1-g)\,d\mu_n < \varepsilon$ for large n. Since $fg \in C_0(\mathbb{R})$,

$$\Big|\int f\,d\mu_n - \int f\,d\mu\Big| \le \Big|\int fg\,d\mu_n - \int fg\,d\mu\Big| + \|f\|_\infty\Big(\int (1-g)\,d\mu_n + \int (1-g)\,d\mu\Big),$$

whose lim sup is at most $2\|f\|_\infty\varepsilon$.

Step (ii)⇒(iii). If F is continuous at x, then $\mu(\{x\}) = 0 = \mu(\partial(-\infty,x])$, and Portmanteau (v) gives $F_n(x) = \mu_n((-\infty,x]) \to F(x)$.

Step (iii)⇒(iv). F is continuous at a and b, so $\mu((a,b]) = F(b) - F(a) = \lim_n (F_n(b) - F_n(a))$.

Step (iv)⇒(i). $D := \{x \mid \mu(\{x\}) \neq 0\}$ is countable, so $\mathbb{R}\setminus D$ is dense. Let $f \in C_0(\mathbb{R})$; it is uniformly continuous, so for $\varepsilon > 0$ there is δ with $|f(x) - f(y)| \leq \varepsilon$ if $|x - y| \leq \delta$. Choose $c_0 < \cdots < c_m$ in $\mathbb{R}\setminus D$ with $\operatorname{supp} f \subset (c_0, c_m]$ and $c_k - c_{k-1} \leq \delta$, and $g := \sum_{k=1}^{m} f(c_{k-1})\mathbb{1}_{(c_{k-1},c_k]}$. Then $\|f - g\|_\infty \leq \varepsilon$ and, as in the Portmanteau proof,

$$\Big|\int f\,d\mu - \int f\,d\mu_n\Big| \leq 2\varepsilon + \sum_k |f(c_{k-1})||\mu((c_{k-1}, c_k]) - \mu_n((c_{k-1}, c_k])| \to 2\varepsilon.$$

■

3.8 Joint distribution and convolution

Motivation. For i.i.d. $X_i \in \mathcal{L}^1$, Kolmogorov's law and the DCT give $\mathbb{E}[f(S_n/n)] \to f(\mathbb{E}[X_1])$ for $f \in C_b$, i.e. the law of S_n/n converges weakly to $\delta_{\mathbb{E}[X_1]}$ (for $X_i \in \mathcal{L}^2$ simply because $\operatorname{Var}(S_n/n) \to 0$). The interesting scale is $S_n/\sqrt{n}$, whose variance stays fixed, and the central limit theorem will identify its limit law as normal. To compute laws of sums we need joint distributions, and for independent summands the law of the sum is a convolution.

Definition 3.38 (Joint distribution)

For real r.v.'s $X_1, \ldots, X_n$ on $(\Omega, \mathcal{A}, P)$, the law $P_{\bar X} := P \circ \bar X^{-1}$ of $\bar X := (X_1, \ldots, X_n) : \Omega \to \mathbb{R}^n$ is the **joint distribution** of $X_1, \ldots, X_n$.

Remark 3.39

(i) $P_{\bar X}(A) = P(\bar X \in A)$, $A \in \mathcal{B}(\mathbb{R}^n)$, is a probability measure (Proposition 2.13), because $\bar X$ is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable. Indeed

$$\mathcal{B}(\mathbb{R}^n) = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R}) = \sigma(\{A_1 \times \cdots \times A_n \mid A_i = (-\infty, x_i]\}),$$

and $\bar X^{-1}(A_1 \times \cdots \times A_n) = \bigcap_i \{X_i \in A_i\} \in \mathcal{A}$ (Remark 1.32). [Supplement] *The equalities:* every open set of $\mathbb{R}^n$ is a countable union of open boxes (Remark 1.27(i)), and boxes are Borel; and $\sigma(\{(-\infty, x]\}) = \mathcal{B}(\mathbb{R})$ in each coordinate, with products of half-open intervals obtained from orthants by finite differences.

(ii) By Theorem 1.9, $P_{\bar X}$ is determined by $P_{\bar X}(A_1 \times \cdots \times A_n) = P(X_1 \in A_1, \ldots, X_n \in A_n)$, even with $A_i = (-\infty, x_i]$.

Remark 3.40 (Transformation theorem)

For Borel $\Phi : \mathbb{R}^n \to \mathbb{R}$ with $\Phi(\bar X) \in \mathcal{L}^1$ or $\Phi \geq 0$: $\mathbb{E}[\Phi(\bar X)] = \int_\Omega \Phi(\bar X)\,dP = \int_{\mathbb{R}^n} \Phi\,dP_{\bar X}$ (same proof as in Remark 2.15).

Example 3.41

(i) Let X, Y be uniform on $[0,1]$. If they are independent, their joint law is uniform on $[0,1]^2$ (Theorem 3.42). If $X = Y$, it is the "uniform distribution on the diagonal", the image of $\lambda|_{[0,1]}$ under $x \mapsto (x, x)$. (ii) For independent $X, Y \sim N(m, \sigma^2)$, Theorem 3.42 gives the joint density $f(x,y) = \frac{1}{2\pi\sigma^2} \exp\big(-\frac{(x-m)^2+(y-m)^2}{2\sigma^2}\big)$, a two-dimensional normal distribution.

Theorem 3.42

Let $X_1, \dots, X_n$ have laws $\mu_1, \dots, \mu_n$ and joint law $P_{\bar{X}}$. Then $X_1, \dots, X_n$ are independent $\iff$ $P_{\bar{X}} = \bigotimes_i \mu_i$, i.e. $P_{\bar{X}}(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$. In this case:

(i) $P_{\bar{X}}$ is determined by $\mu_1, \dots, \mu_n$;

(ii) for every permutation π and $\phi \geq 0$ (or $P_{\bar{X}}$-integrable), $\int \phi \, dP_{\bar{X}}$ equals the iterated integral in the order π;

(iii) if $\mu_i(dx_i) = f_i(x_i)\, dx_i$, then $P_{\bar{X}}(d\bar{x}) = \bar{f}(\bar{x})\, d\bar{x}$ with $\bar{f}(\bar{x}) = \prod_i f_i(x_i)$.

Proof. The equivalence is Definition 3.18 rewritten, since $\mu_i(A_i) = P(X_i \in A_i)$.

(i) Theorem 3.30.

(ii) Theorem 3.31 and Remark 3.32.

(iii) $\bar{f} \geq 0$ is Borel with $\int \bar{f}\, d\bar{x} = \prod_i \int f_i = 1$ (Fubini for Lebesgue measure, which is the product of one-dimensional Lebesgue measures on boxes and hence everywhere by Theorem 1.9), so $\bar{\mu}(A) := \int_A \bar{f}\, d\bar{x}$ is a probability measure. On rectangles, $\bar{\mu}(A_1 \times \cdots \times A_n) = \prod_i \int_{A_i} f_i = \prod_i \mu_i(A_i) = P_{\bar{X}}(A_1 \times \cdots \times A_n)$, so $\bar{\mu} = P_{\bar{X}}$ by Theorem 1.9. ∎

For independent $X_1, \dots, X_n$, how is the law of $S_n = X_1 + \cdots + X_n$ obtained from the laws of the X_i? Let $T_x(y) := x + y$ be translation by x.

Theorem 3.43 (Convolution)

Let X_1, X_2 be independent with laws μ_1, μ_2.

(i) $P_{X_1+X_2} = \mu_1 * \mu_2 := \int \mu_2 \circ T_{x_1}^{-1}(\cdot)\, \mu_1(dx_1)$, i.e. for $A \in \mathcal{B}(\mathbb{R})$

$$\mu_1 * \mu_2(A) = \iint \mathbb{1}_A(x_1 + x_2)\, \mu_1(dx_1)\, \mu_2(dx_2) = \int \mu_2(A - x_1)\, \mu_1(dx_1),$$

where $A - x_1 := \{x - x_1 \mid x \in A\}$.

(ii) If μ_2 has density f_2, then $\mu_1 * \mu_2$ has density $f(x) = \int f_2(x - x_1)\, \mu_1(dx_1)$, which equals $(f_1 * f_2)(x) := \int f_1(x_1) f_2(x - x_1)\, dx_1$ if $\mu_1(dx_1) = f_1(x_1)\, dx_1$.

Proof. (i) With $\bar{A} := \{(x_1, x_2) \mid x_1 + x_2 \in A\} \in \mathcal{B}(\mathbb{R}^2)$ and Theorem 3.42,

$$P(X_1 + X_2 \in A) = (\mu_1 \otimes \mu_2)(\bar{A}) = \int \Big(\int \mathbb{1}_{A - x_1}(x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1) = \int \mu_2(A - x_1)\, \mu_1(dx_1).$$

(ii) Substituting $x = x_1 + x_2$ and using Fubini (Theorem 3.42(ii)):

$$\begin{aligned}(\mu_1 * \mu_2)(A) &= \int \int_{A - x_1} f_2(x_2)\, dx_2\, \mu_1(dx_1) = \int \int_A f_2(x - x_1)\, dx\, \mu_1(dx_1) \\ &= \int_A \Big(\int f_2(x - x_1)\, \mu_1(dx_1) \Big) dx.\end{aligned}$$

∎

Example 3.44

(i) If $X_i \sim N(m_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(m_1 + m_2, \sigma_1^2 + \sigma_2^2)$, i.e. $f_{m_1,\sigma_1^2} * f_{m_2,\sigma_2^2} = f_{m_1+m_2,\sigma_1^2+\sigma_2^2}$. [Supplement] After shifting, let $m_i = 0$, $a = \sigma_1^2$, $b = \sigma_2^2$. Completing

the square in y,

$$\int e^{-\frac{y^2}{2a}-\frac{(x-y)^2}{2b}}\,dy = e^{-\frac{x^2}{2(a+b)}}\int e^{-\frac{a+b}{2ab}(y-\frac{ax}{a+b})^2}\,dy = e^{-\frac{x^2}{2(a+b)}}\sqrt{\frac{2\pi ab}{a+b}},$$

and multiplying by $\frac{1}{2\pi\sqrt{ab}}$ gives $f_{0,a+b}(x)$.
(ii) The **Gamma distribution** $\Gamma_{\alpha,p}$ has density $\gamma_{\alpha,p}(x) = \frac{\alpha^p}{\Gamma(p)}x^{p-1}e^{-\alpha x}$ for $x > 0$ (0 otherwise). If $X_i \sim \Gamma_{\alpha,p_i}$ are independent, then $X_1 + X_2 \sim \Gamma_{\alpha,p_1+p_2}$. [Supplement] For $x > 0$, substituting $y = xt$,

$$\begin{aligned}(\gamma_{\alpha,p_1} * \gamma_{\alpha,p_2})(x) &= \frac{\alpha^{p_1+p_2}e^{-\alpha x}}{\Gamma(p_1)\Gamma(p_2)}\int_0^x y^{p_1-1}(x-y)^{p_2-1}dy \\ &= \frac{\alpha^{p_1+p_2}e^{-\alpha x}x^{p_1+p_2-1}}{\Gamma(p_1)\Gamma(p_2)}B(p_1,p_2) = \gamma_{\alpha,p_1+p_2}(x),\end{aligned}$$

using $B(p_1,p_2) = \Gamma(p_1)\Gamma(p_2)/\Gamma(p_1+p_2)$. For $p_i = 1$: a sum $S_n = T_1 + \cdots + T_n$ of independent $\mathrm{Exp}(\alpha)$ variables has the Gamma density $\gamma_{\alpha,n}(x) = \frac{\alpha^n}{(n-1)!}x^{n-1}e^{-\alpha x}$, $x > 0$.

Example 3.45 (The waiting time paradox)

Let $T_1, T_2, \ldots$ be independent $\mathrm{Exp}(\alpha)$ waiting times (between calls at a call centre, or buses at a stop), so $\mathbb{E}[T_i] = \int_0^\infty x\alpha e^{-\alpha x}dx = \frac{1}{\alpha}$. Let $S_n := T_1 + \cdots + T_n$, $S_0 := 0$, fix $t > 0$, and let X be the time from the last event before t to t, and Y the time from t to the next event: $X = t - S_n$, $Y = S_{n+1} - t$ on $\{S_n \le t < S_{n+1}\}$.
Question: what is $\mathbb{E}[Y]$? The naive guess $\frac{1}{2\alpha}$ is wrong:

(i) $Y \sim \mathrm{Exp}(\alpha)$, so $\mathbb{E}[Y] = \frac{1}{\alpha}$;

(ii) X is $\mathrm{Exp}(\alpha)$ "compressed to $[0,t]$": $P(X \ge s) = e^{-\alpha s}$ for $0 \le s \le t$ and $P(X = t) = e^{-\alpha t}$, so $\mathbb{E}[X] = \int_0^t s\alpha e^{-\alpha s}ds + te^{-\alpha t} = \frac{1}{\alpha}(1 - e^{-\alpha t})$;

(iii) X and Y are independent.

The interval $[t-X, t+Y]$ containing t has mean length $\frac{1}{\alpha}(2 - e^{-\alpha t}) \approx \frac{2}{\alpha}$: a fixed time is more likely to fall into a long gap.
Proof. For $0 \le x \le t$, $y \ge 0$, decompose according to the number n of events in $[0,t]$; S_n and T_{n+1} are independent with densities $\gamma_{\alpha,n}$ and $\alpha e^{-\alpha r}$:

$$\begin{aligned}P(X \ge x, Y \ge y) &= P(T_1 \ge t + y) + \sum_{n\ge 1} P(S_n \le t - x,\ T_{n+1} \ge t + y - S_n) \\ &= e^{-\alpha(t+y)} + \sum_{n\ge 1}\int_0^{t-x} \gamma_{\alpha,n}(s)e^{-\alpha(t+y-s)}\,ds \\ &= e^{-\alpha(t+y)}\Big(1 + \int_0^{t-x} e^{\alpha s}\sum_{n\ge 1}\gamma_{\alpha,n}(s)\,ds\Big).\end{aligned}$$

Since $\sum_{n\ge1}\gamma_{\alpha,n}(s) = \alpha e^{-\alpha s}\sum_{n\ge1}\frac{(\alpha s)^{n-1}}{(n-1)!} = \alpha$, the bracket is $1 + (e^{\alpha(t-x)} - 1) = e^{\alpha(t-x)}$, and $P(X \ge x, Y \ge y) = e^{-\alpha y}e^{-\alpha x}$. Setting $x = 0$ gives (i), $y = 0$ gives (ii), and the product form gives (iii).

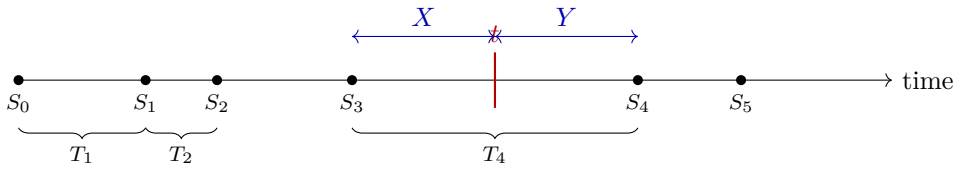


Figure. *The waiting time paradox. The gap containing the fixed time t (here T_4) is size-biased: on average it is about twice as long as a typical gap, and the residual waiting time Y has the same law as a full gap.*

3.9 Characteristic functions

Motivation. Convolution of laws is awkward; multiplication is easy. The characteristic function (Fourier transform of a law) turns sums of independent variables into products, determines the law uniquely, and turns weak convergence into pointwise convergence (Lévy). This is the machinery behind the central limit theorem.

Let $\langle \cdot, \cdot \rangle$ be the inner product on $\mathbb{R}^n$ and $\mathcal{M}_+^1(\mathbb{R}^n)$ the probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ the **characteristic function** is

$$\hat{\mu}(u) := \int e^{i\langle u,y\rangle} \mu(dy) := \int \cos\langle u,y\rangle \, \mu(dy) + i \int \sin\langle u,y\rangle \, \mu(dy), \qquad u \in \mathbb{R}^n.$$

Theorem 3.46

For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$: (i) $|\hat{\mu}(u)| \le \hat{\mu}(0) = 1$; (ii) $\hat{\mu}(-u) = \overline{\hat{\mu}(u)}$; (iii) $\hat{\mu}$ is uniformly continuous; (iv) $\hat{\mu}$ is positive definite: $\sum_{j,k=1}^m c_j \bar{c}_k \hat{\mu}(u_j - u_k) \ge 0$ for $c_j \in \mathbb{C}$, $u_j \in \mathbb{R}^n$.

Proof [Supplement]. (i) $|\hat{\mu}(u)| \le \int |e^{i\langle u,y\rangle}| \, d\mu = 1 = \hat{\mu}(0)$.

(ii) Conjugate the integrand.

(iii) $|\hat{\mu}(u+h) - \hat{\mu}(u)| \le \int |e^{i\langle h,y\rangle} - 1| \, \mu(dy)$, which does not depend on u and tends to 0 as $h \to 0$ by the DCT.

(iv) $\sum_{j,k} c_j \bar{c}_k \hat{\mu}(u_j - u_k) = \int \left| \sum_j c_j e^{i\langle u_j, y\rangle} \right|^2 \mu(dy) \ge 0$. ∎

Theorem 3.47 (Uniqueness theorem)

If $\mu_1, \mu_2 \in \mathcal{M}_+^1(\mathbb{R}^n)$ and $\hat{\mu}_1 = \hat{\mu}_2$, then $\mu_1 = \mu_2$.

Proof [Supplement] (Gaussian smoothing).

Step 1 (a Fourier formula for the Gaussian). Let $p_\sigma(x) := (2\pi\sigma^2)^{-n/2} e^{-\|x\|^2/(2\sigma^2)}$. Substituting $y = \sigma u$ in Theorem 3.52 below (whose proof uses nothing from this section), with u replaced by x/σ, gives

$$p_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \, du.$$

Step 2 (smoothed measure is determined by $\hat{\mu}$). For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ let $h_\sigma(x) := \int p_\sigma(x-y) \, \mu(dy)$. By Step 1 and Fubini (the integrand is bounded by $e^{-\sigma^2\|u\|^2/2}$, integrable w.r.t. $du \otimes \mu$),

$$h_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \Big(\int e^{-i\langle y,u\rangle} \mu(dy) \Big) du = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \hat{\mu}(-u) \, du,$$

which depends on μ only through $\hat{\mu}$.

Step 3 (letting $\sigma \to 0$). For $f \in C_b(\mathbb{R}^n)$, with γ the standard Gaussian on $\mathbb{R}^n$ and the substitution $x = y + \sigma z$,

$$\int f(x) h_\sigma(x)\,dx = \iint f(y+\sigma z)\,\gamma(dz)\,\mu(dy) \xrightarrow[\sigma\to 0]{} \int f\,d\mu$$

by the DCT. By Step 2 the left side is the same for μ_1 and μ_2, so $\int f\,d\mu_1 = \int f\,d\mu_2$ for all $f \in C_b(\mathbb{R}^n)$, and $\mu_1 = \mu_2$ by Example 2.30. ∎

Theorem 3.48 (Bochner)

If $\phi : \mathbb{R}^n \to \mathbb{C}$ is continuous, positive definite and $\phi(0) = 1$, there is exactly one $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat\mu = \phi$.

On the proof

Uniqueness is Theorem 3.47. Existence is not proved in the lectures; see Klenke, *Probability Theory*, Thm. 15.29. It is not used below.

Theorem 3.49 (Lévy's continuity theorem)

Let $\mu_m \in \mathcal{M}_+^1(\mathbb{R}^n)$.

(i) If $\mu_m \to \mu$ weakly, then $\hat\mu_m \to \hat\mu$ uniformly on compact sets.

(ii) If $\hat\mu_m \to \phi$ pointwise for some $\phi : \mathbb{R}^n \to \mathbb{C}$ continuous at 0, then there is a unique $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat\mu = \phi$, and $\mu_m \to \mu$ weakly.

Proof [Supplement] (part (ii) for $n = 1$, the case used in the CLT).

Step (i) Tightness. Given $\varepsilon > 0$ choose R with $\mu(\{\|y\| \ge R\}) < \varepsilon$. By Portmanteau (iii) for the closed set $\{\|y\| \ge R\}$, $\mu_m(\{\|y\| \ge R\}) < \varepsilon$ for $m \ge m_0$; enlarging R covers the finitely many $m < m_0$.

Step (i) Equicontinuity. Then $|\hat\mu_m(u) - \hat\mu_m(v)| \le \int_{\{\|y\|<R\}} |e^{i\langle u-v,y\rangle} - 1|\,d\mu_m + 2\varepsilon \le R\|u-v\| + 2\varepsilon$ for all m. Pointwise convergence ($\cos, \sin \in C_b$) plus this uniform equicontinuity gives uniform convergence on compacts, by covering a compact set with finitely many small balls.

Step (ii) Tightness from ϕ. For $\delta > 0$ and $\nu \in \mathcal{M}_+^1(\mathbb{R})$, Fubini gives

$$\frac{1}{\delta}\int_{-\delta}^{\delta}(1-\hat\nu(u))\,du = 2\int\Big(1 - \frac{\sin \delta x}{\delta x}\Big)\nu(dx) \ge \nu(\{|x| \ge 2/\delta\}),$$

since $1 - \frac{\sin s}{s} \ge \frac12$ for $|s| \ge 2$. As $\phi(0) = \lim \hat\mu_m(0) = 1$ and ϕ is continuous at 0, choose δ with $\frac{1}{\delta}\int_{-\delta}^{\delta}|1-\phi|\,du < \varepsilon$. By the DCT ($|\hat\mu_m| \le 1$), $\frac{1}{\delta}\int_{-\delta}^{\delta}(1-\hat\mu_m) \to \frac{1}{\delta}\int_{-\delta}^{\delta}(1-\phi)$. Hence $\mu_m(\{|x| \ge 2/\delta\}) < 2\varepsilon$ for large m: (μ_m) is tight.

Step (ii) Helly selection. Let F_m be the distribution functions. A diagonal argument yields a subsequence (m_k) along which $F_{m_k}(q) \to H(q)$ for every rational q. Put $G(x) := \inf_{q>x} H(q)$; G is increasing and right continuous. If G is continuous at x and $q < x < q'$ are rational, then $H(q) \le \liminf_k F_{m_k}(x) \le \limsup_k F_{m_k}(x) \le H(q')$. Letting $q \uparrow x$, $q' \downarrow x$ gives $F_{m_k}(x) \to G(x)$. Tightness forces $G(-\infty) = 0$ and $G(+\infty) = 1$, so G is the distribution function of some ν (Remark 2.15), and $\mu_{m_k} \to \nu$ weakly by Corollary 3.37.

Step (ii) Identification. By (i), $\hat\nu = \lim_k \hat\mu_{m_k} = \phi$. Every subsequence of (μ_m) is again tight, so it has a further subsequence converging weakly to some ν' with $\hat\nu' = \phi$; by Theorem 3.47, $\nu' = \nu =: \mu$. If $\mu_m \not\to \mu$ weakly, there would be $f \in C_b$, $\varepsilon > 0$ and a subsequence with $|\int f\,d\mu_m - \int f\,d\mu| \ge \varepsilon$, contradicting the above.

For $n \geq 2$ the same argument works with multivariate distribution functions in Helly's theorem; we do not need it. ■

For an $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable $\bar{X} = (X_1, \dots, X_n)$ on $(\Omega, \mathcal{A}, P)$,

$$\varphi_{\bar{X}}(u) := \hat{P}_{\bar{X}}(u) = \int e^{i\langle u, y\rangle} P_{\bar{X}}(dy) = \mathbb{E}[e^{i\langle u, \bar{X}\rangle}]$$

is the **characteristic function of** $\bar{X}$.

Remark 3.50

$X_1, \dots, X_n$ are independent iff $\varphi_{\bar{X}}(u_1, \dots, u_n) = \prod_j \varphi_{X_j}(u_j)$ for all $u \in \mathbb{R}^n$. "⇒" is Theorem 3.20 (applied to real and imaginary parts, via Corollary 3.21). For "⇐", the right side is the characteristic function of $\bigotimes_j P_{X_j}$ (by Fubini), so Theorem 3.47 gives $P_{\bar{X}} = \bigotimes_j P_{X_j}$, and Theorem 3.42 applies.

Theorem 3.51

If $X_1, \dots, X_n$ are independent, $\alpha \in \mathbb{R}$ and $S := \alpha \sum_k X_k$, then $\varphi_S(u) = \prod_{k=1}^n \varphi_{X_k}(\alpha u)$.

Proof. $\varphi_S(u) = \mathbb{E}[\prod_k e^{i\alpha u X_k}] = \prod_k \mathbb{E}[e^{i\alpha u X_k}]$ by independence (Corollary 3.21, applied to the bounded complex factors via real and imaginary parts and induction). ■

Theorem 3.52 (Gaussian characteristic function)

For all $u \in \mathbb{R}^n$: $(2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\langle u, y\rangle} e^{-\frac{1}{2}\|y\|^2} dy = e^{-\frac{1}{2}\|u\|^2}$.

Proof [Supplement]. By Fubini the integral factorises over coordinates, so let $n = 1$ and $\varphi(u) := (2\pi)^{-1/2} \int e^{iuy} e^{-y^2/2} dy$. Differentiating under the integral (dominated by $|y| e^{-y^2/2}$) and integrating by parts,

$$\varphi'(u) = (2\pi)^{-1/2} \int iy\, e^{iuy} e^{-y^2/2} dy = (2\pi)^{-1/2} \int i\, e^{iuy} \frac{d}{dy}\big(-e^{-y^2/2}\big) dy = -u\varphi(u).$$

With $\varphi(0) = 1$ this gives $\varphi(u) = e^{-u^2/2}$. ■

Example 3.53

(i) $\hat{\delta}_a(u) = e^{iua}$. (ii) For $\mu = \sum_i \alpha_i \delta_{a_i}$ ($\alpha_i \geq 0$, $\sum \alpha_i = 1$), $\hat{\mu}(u) = \sum_i \alpha_i e^{iua_i}$. In particular the binomial distribution $\beta_n^p = \sum_k \binom{n}{k} p^k q^{n-k} \delta_k$ ($q = 1 - p$) has $\hat{\beta}_n^p(u) = (q + pe^{iu})^n$, and the Poisson distribution $\pi_\alpha = \sum_n e^{-\alpha} \frac{\alpha^n}{n!} \delta_n$ has $\hat{\pi}_\alpha(u) = e^{-\alpha} \sum_n \frac{(\alpha e^{iu})^n}{n!} = e^{\alpha(e^{iu} - 1)}$.

3.10 Central limit theorem

Motivation. After centring and scaling by the standard deviation, a sum of many independent, individually negligible contributions is approximately standard normal, whatever their laws. Lindeberg's condition makes "individually negligible" precise. The proof compares $\prod_k \varphi_{X_k}(u/s_n)$ with $\exp(\sum_k (\varphi_{X_k}(u/s_n) - 1))$ using a second-order Taylor expansion.

Definition 3.54 (Central limit property)

Let $X_1, X_2, \cdots \in \mathcal{L}^2$ be independent with positive variances, $S_n := X_1 + \cdots + X_n$ and $S_n^* := \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\mathrm{Var}(S_n)}}$ (so $\mathbb{E}[S_n^*] = 0$, $\mathrm{Var}(S_n^*) = 1$). (X_n) has the **central limit property** (CLP) if $P_{S_n^*} \to N(0,1)$ weakly, equivalently (Corollary 3.37)

$$\lim_n P(S_n^* \le b) = \Phi(b) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{b} e^{-x^2/2} dx \qquad \forall b \in \mathbb{R}.$$

Theorem 3.55 (Central limit theorem, Lindeberg)

Let $X_1, X_2, \cdots \in \mathcal{L}^2$ be independent, $\sigma_n^2 := \mathrm{Var}(X_n) > 0$, $s_n := (\sum_{k \le n} \sigma_k^2)^{1/2}$, and assume **Lindeberg's condition**

$$L_n(\varepsilon) := \sum_{k=1}^{n} \int_{\{|X_k - \mathbb{E}[X_k]| \ge \varepsilon s_n\}} \Big(\frac{X_k - \mathbb{E}[X_k]}{s_n} \Big)^2 dP \xrightarrow[n\to\infty]{} 0 \qquad \forall \varepsilon > 0. \tag{L}$$

Then (X_n) has the CLP.

Remark 3.56

(i) i.i.d. sequences in $\mathcal{L}^2$ satisfy (L): with $m = \mathbb{E}[X_1]$, $\sigma^2 = \mathrm{Var}(X_1)$, $s_n^2 = n\sigma^2$ and $L_n(\varepsilon) = \sigma^{-2} \int_{\{|X_1 - m| \ge \varepsilon \sqrt{n}\sigma\}} (X_1 - m)^2 dP \to 0$ by the DCT.

(ii) **Lyapunov's condition** (often with $\delta = 1$) implies (L):

$$\exists \delta > 0 : \quad \lim_n \sum_{k=1}^{n} \frac{\mathbb{E}|X_k - \mathbb{E}[X_k]|^{2+\delta}}{s_n^{2+\delta}} = 0. \tag{Lya}$$

Indeed $|X_k - \mathbb{E}X_k| \ge \varepsilon s_n$ implies $|X_k - \mathbb{E}X_k|^2 \le |X_k - \mathbb{E}X_k|^{2+\delta} / (\varepsilon s_n)^{\delta}$, so $L_n(\varepsilon) \le \varepsilon^{-\delta} \sum_k \mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta} / s_n^{2+\delta}$.

(iii) Uniformly bounded X_n with $s_n \to \infty$ satisfy (Lya) for every $\delta > 0$: if $|X_k| \le \alpha/2$, then $|X_k - \mathbb{E}X_k| \le \alpha$ and

$$\sum_k \frac{\mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta}}{s_n^{2+\delta}} \le \sum_k \frac{\alpha^{\delta} \mathbb{E}|X_k - \mathbb{E}X_k|^2}{s_n^2 s_n^{\delta}} = \Big(\frac{\alpha}{s_n} \Big)^{\delta} \to 0.$$

Lemma 3.57

Lindeberg's condition implies

$$\lim_n \max_{1 \le k \le n} \frac{\sigma_k}{s_n} = 0. \tag{3.14}$$

Proof. For $k \le n$ and $\varepsilon > 0$, splitting the integral on $\{|X_k - \mathbb{E}X_k| \ge \varepsilon s_n\}$:

$$\Big(\frac{\sigma_k}{s_n} \Big)^2 = \int \Big(\frac{X_k - \mathbb{E}X_k}{s_n} \Big)^2 dP \le L_n(\varepsilon) + \varepsilon^2.$$

■

Lemma 3.58

For $t \in \mathbb{R}$, $n \in \mathbb{N}$: $\left|e^{it} - \sum_{k=0}^{n-1} \frac{(it)^k}{k!}\right| \le \frac{|t|^n}{n!}$.

Proof. Taylor's formula with integral remainder for e^{it} (real and imaginary parts) gives the remainder $R_n(t) = \frac{1}{(n-1)!}\int_0^t (t-s)^{n-1} i^n e^{is} ds$, and $|R_n(t)| \le \frac{1}{(n-1)!}\int_0^{|t|} s^{n-1} ds = \frac{|t|^n}{n!}$. ∎

Theorem 3.59

For $X \in \mathcal{L}^2$, φ_X is twice continuously differentiable with $\varphi_X'(u) = i\int X e^{iuX} dP$, $\varphi_X''(u) = -\int X^2 e^{iuX} dP$; in particular $\varphi_X'(0) = i\mathbb{E}[X]$, $\varphi_X''(0) = -\mathbb{E}[X^2]$, $|\varphi_X''| \le \mathbb{E}[X^2]$. Moreover

$$\varphi_X(u) = 1 + iu\mathbb{E}[X] + \tfrac{1}{2}\theta(u)u^2\mathbb{E}[X^2] \qquad \text{with } \theta(u) \in \mathbb{C},\ |\theta(u)| \le 1.$$

Proof.

Step 1 (derivatives). $\frac{d}{du}e^{iuX} = iXe^{iuX}$, $\frac{d^2}{du^2}e^{iuX} = -X^2e^{iuX}$. By Lemma 3.58 with $n = 1$, the difference quotients are dominated by $|X|$, resp. X^2, which are integrable. The DCT allows differentiation under the integral and gives continuity of the derivatives.

Step 2 (expansion). Lemma 3.58 with $n = 2$, $t = uX$: $|e^{iuX} - 1 - iuX| \le \frac{1}{2}u^2X^2$, so $|\varphi_X(u) - 1 - iu\mathbb{E}[X]| \le \frac{1}{2}u^2\mathbb{E}[X^2]$. Put $\theta(u) := 0$ if $u^2\mathbb{E}[X^2] = 0$ and $\theta(u) := \frac{\varphi_X(u) - 1 - iu\mathbb{E}[X]}{\frac{1}{2}u^2\mathbb{E}[X^2]}$ otherwise. ∎

From now on $X_1, X_2, \dots \in \mathcal{L}^2$ are independent with $\mathbb{E}[X_n] = 0$, $\sigma_n^2 = \mathrm{Var}(X_n) > 0$, $s_n = (\sum_{k\le n}\sigma_k^2)^{1/2}$.

Theorem 3.60

Assume

$$\text{(F)}\quad \lim_n \max_{k\le n} \frac{\sigma_k}{s_n} = 0 \ \text{(Feller)}, \qquad \text{(b)}\quad \lim_n \sum_{k=1}^n \left(\varphi_{X_k}\Big(\frac{u}{s_n}\Big) - 1\right) = -\frac{u^2}{2} \quad \forall u \in \mathbb{R}.$$

Then (X_n) has the CLP.

Proof.

Step 1 (reduction). By Theorem 3.51, $\varphi_{S_n^*}(u) = \prod_{k\le n}\varphi_{X_k}(u/s_n)$. If

$$\lim_n \prod_{k=1}^n \varphi_{X_k}\Big(\frac{u}{s_n}\Big) = e^{-u^2/2} \qquad \forall u, \tag{3.15}$$

then $\varphi_{S_n^*} \to e^{-u^2/2} = \widehat{N(0,1)}$ (Theorem 3.52), which is continuous at 0. Lévy's theorem 3.49 and Theorem 3.47 give $P_{S_n^*} \to N(0,1)$ weakly.

Step 2 (comparison with exponentials). By (b), $\prod_k \exp(\varphi_{X_k}(u/s_n) - 1) = \exp\big(\sum_k(\varphi_{X_k}(u/s_n) - 1)\big) \to e^{-u^2/2}$, so it suffices that the two products differ by $o(1)$. Since $|\varphi_{X_k}| \le 1$, $|\exp(\varphi_{X_k} - 1)| = \exp(\mathrm{Re}\,\varphi_{X_k} - 1) \le 1$. For a_k, b_k in the closed unit disc, telescoping gives $|\prod_k a_k - \prod_k b_k| \le \sum_k |a_k - b_k|$. Hence, with $z_k := \varphi_{X_k}(u/s_n) - 1$,

$$\Big|\prod_k \varphi_{X_k}\Big(\frac{u}{s_n}\Big) - \prod_k e^{z_k}\Big| \le D_n := \sum_{k=1}^n |z_k + 1 - e^{z_k}|.$$

Step 3. Fix $\varepsilon \in (0, \frac{1}{2}]$. For $|z| \le \varepsilon$, $|z + 1 - e^z| \le \sum_{j\ge 2}|z|^j/j! \le |z|^2 \le \varepsilon|z|$. By Theorem 3.59 with

$\mathbb{E}[X_k]=0$, $|z_k|\le \frac{1}{2}\frac{u^2}{s_n^2}\sigma_k^2$, and by (F) this is $<\varepsilon$ for all $k\le n$ once $n\ge n_0$. Then

$$D_n \le \varepsilon\sum_k |z_k| \le \varepsilon\frac{u^2}{2}\sum_k \frac{\sigma_k^2}{s_n^2} = \varepsilon\frac{u^2}{2},$$

so $D_n\to 0$. ∎

Proof of Theorem 3.55.

Step 1. We may assume $\mathbb{E}[X_n]=0$. By Lemma 3.57, (L)⇒(F), so by Theorem 3.60 it remains to derive (b) from (L).

Step 2 (pointwise bounds). Fix u and let

$$Y_k := \Big|e^{i\frac{u}{s_n}X_k} - 1 - i\frac{u}{s_n}X_k + \frac{1}{2}\frac{u^2}{s_n^2}X_k^2\Big|.$$

Lemma 3.58 ($n=3$) gives $Y_k\le \frac{1}{6}|\frac{u}{s_n}X_k|^3$, and the triangle inequality with Lemma 3.58 ($n=2$) gives $Y_k\le \frac{u^2}{s_n^2}X_k^2$.

Step 3. Since $\mathbb{E}[X_k]=0$ and $\sum_k \sigma_k^2/s_n^2=1$,

$$\Big|\sum_{k=1}^n\Big(\varphi_{X_k}\Big(\frac{u}{s_n}\Big)-1\Big)+\frac{u^2}{2}\Big| = \Big|\sum_k \mathbb{E}\Big[e^{i\frac{u}{s_n}X_k}-1-i\frac{u}{s_n}X_k+\frac{u^2}{2s_n^2}X_k^2\Big]\Big| \le \sum_k \mathbb{E}[Y_k].$$

Step 4. For $\varepsilon>0$, use the second bound on $\{|X_k|\ge \varepsilon s_n\}$ and the first on the complement, where $\frac{1}{s_n^3}\int_{\{|X_k|<\varepsilon s_n\}}|X_k|^3dP \le \frac{\varepsilon}{s_n^2}\sigma_k^2$:

$$\sum_k \mathbb{E}[Y_k] \le u^2\sum_k\int_{\{|X_k/s_n|\ge\varepsilon\}}\Big(\frac{X_k}{s_n}\Big)^2dP + \frac{|u|^3}{6}\varepsilon\sum_k\frac{\sigma_k^2}{s_n^2} = u^2L_n(\varepsilon)+\frac{|u|^3}{6}\varepsilon.$$

Hence $\limsup_n \sum_k \mathbb{E}[Y_k]\le \frac{|u|^3}{6}\varepsilon$ for every $\varepsilon>0$, which proves (b). ∎

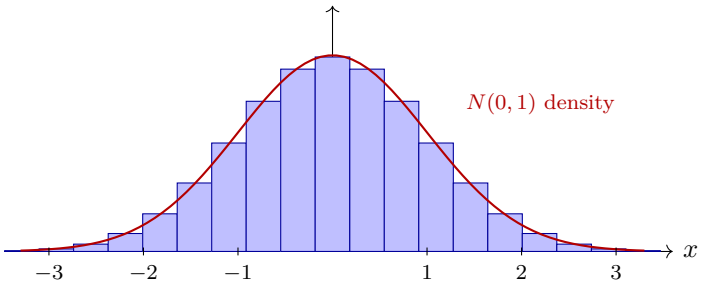


Figure. *The CLT for coin tossing: the law of the standardised number of heads S_n^* for $n=30$, $p=\frac{1}{2}$, drawn as bars of area $P(S_n=k)$ and width $1/s_n$, against the standard normal density.*

Example 3.61 (Applications)

(i) Ruin probability. An insurer holds n contracts with i.i.d. claims $X_i\ge 0$ in $\mathcal{L}^2$, $m=\mathbb{E}[X_1]$, $\sigma^2=\mathrm{Var}(X_1)$, and charges the premium $\Pi := m+\lambda\sigma^2$ (average claim plus safety loading, $\lambda>0$). With initial capital K, no interest, and payments at the end of the period, ruin is $R:=\{S_n>K+n\Pi\}$. With $S_n^*=\frac{S_n-nm}{\sqrt{n}\sigma}$ the CLT gives

$$P(R)=P\Big(S_n^*>\frac{K+n\lambda\sigma^2}{\sqrt{n}\sigma}\Big)\approx 1-\Phi(c_n),\qquad c_n:=\frac{K+n\lambda\sigma^2}{\sqrt{n}\sigma}\to\infty.$$

As a function of n, c_n increases once $n > K/(\lambda\sigma^2)$, so the normal approximation suggests that the ruin probability eventually decreases as the portfolio grows.

Remark 3.62

Here the normal approximation is applied at levels $c_n \to \infty$, where the CLT alone gives no rate. If moreover $\mathbb{E}|X_1 - m|^3 < \infty$, the Berry–Esseen theorem (Klenke, Thm. 15.51) gives the uniform bound

$$\sup_x |P(S_n^* \le x) - \Phi(x)| \le \frac{C}{\sqrt{n}} \frac{\mathbb{E}|X_1 - m|^3}{\sigma^3},$$

so in particular $|P(S_n^* > c_n) - (1 - \Phi(c_n))| \le \frac{C}{\sqrt{n}} \mathbb{E}|X_1 - m|^3/\sigma^3$, which justifies the numbers below. With $n = 2000$, $\sigma = 60$, $\lambda = 0.5‰$: (a) $K = 0$ gives $P(R) \approx 1 - \Phi(1.342) \approx 9\%$; (b) $K = 1500$ gives $P(R) \approx 3\%$. For $P(R) \le 1‰$ with $K = 0$ one needs $\Phi(\sqrt{n}\lambda\sigma) \ge 0.999$, i.e. $n \ge 10\,611$.

(ii) Stirling's formula via the CLT (Weng 1977). Stirling (1730) proved

$$n! \approx \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}, \tag{3.16}$$

and De Moivre used it for the CLT for Bernoulli trials (1733). Conversely, the CLT (whose proof above did not use Stirling) yields (3.16). Let X_i be i.i.d. Poisson(1); then $S_n \sim \pi_n$ with $\mathbb{E}[S_n] = \mathrm{Var}(S_n) = n$ and $S_n^* = t_n(S_n)$, $t_n(x) = \frac{x-n}{\sqrt{n}}$. For $f_\infty(x) := x^- = (-x) \vee 0$, a telescoping sum gives

$$\int f_\infty \, dP_{S_n^*} = e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} \frac{n-k}{\sqrt{n}} = \frac{e^{-n}}{\sqrt{n}} \Big(n + \sum_{k=1}^{n} \Big(\frac{n^{k+1}}{k!} - \frac{n^k}{(k-1)!} \Big) \Big) = \frac{e^{-n} n^{n+\frac{1}{2}}}{n!},$$

while $\int f_\infty \, dN(0,1) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} (-x) e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}}$. So (3.16) follows once

$$\int f_\infty \, dP_{S_n^*} \to \int f_\infty \, dN(0,1). \tag{3.17}$$

This is not immediate, since f_∞ is unbounded. With $f_m := f_\infty \wedge m \in C_b$, the CLT gives $\int f_m \, dP_{S_n^*} \to \int f_m \, dN(0,1)$. The remainder $g_m := f_\infty - f_m \ge 0$ is uniformly small:

$$\int g_m \, dP_{S_n^*} \le \int_{(-\infty, -m]} |x| \, dP_{S_n^*} \le \frac{1}{m} \int x^2 \, dP_{S_n^*} = \frac{1}{m} \mathrm{Var}(S_n^*) = \frac{1}{m},$$

and likewise $\int g_m \, dN(0,1) \le \frac{1}{m}$. A 3ε-argument gives (3.17).